

Jitesh Rachamadugu

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Professional Summary

AI & Machine Learning Engineer with hands-on experience building **agentic AI workflows**, ML-driven systems, and production-ready APIs using **Python**. Skilled in **model development, automation pipelines, and policy-bound AI agents**, with practical exposure to **Azure-based deployments**. Strong foundations in **DSA and ML engineering**, focused on delivering reliable, **scalable enterprise-grade solutions**.

Education

Gandhi Institute of Technology and Management (GITAM), Hyderabad, India May 2021 – May 2025
Bachelor of Technology in Computer Science — CGPA: **7.86**
Relevant Coursework: Data Structures, Algorithms, Object-Oriented Design, Machine Learning, Database Systems, Operating Systems

Professional Experience

Software Engineer Intern – AI Solutions Jan 2025 – Mar 2025
HSBC, Pune, India

- Built and deployed an **LLM-powered agentic chatbot** using **FastAPI** and Azure, designed with **secure prompt boundaries, output validation, and auditability**, reducing customer query resolution time by **40%**.
- Fine-tuned Falcon-7B** with **LoRA adapters** on 10k+ transcripts, embedding compliance and policy-alignment layers to reduce non-compliant outputs by **30%**.
- Engineered advanced **prompt optimization strategies (COT, Zero/Few-Shot)** to improve contextual accuracy, reduce hallucinations, and ensure **policy-aligned, security-relevant outputs**.
- Optimized **Azure DevOps CI/CD pipelines** with automated rollbacks and predictive failure modeling, enhancing **deployment reliability and uptime**.

Artificial Intelligence Research Intern May 2024 – Sep 2024
IIITDM Kurnool, India

- Built a **hybrid ML pathfinding algorithm** for ROS robots, improving navigation accuracy by **20%**.
- Optimized **sensor-data pipelines** using C++ and performed large-scale EDA and drift analysis across 1,000+ GPU simulations.
- Built a **C++/Python framework** for model validation, reducing approval cycles by **70%** and improving system reliability.

Software Projects

Enterprise-Scale RAG Pipeline for Aerospace Systems May 2025 – July 2025

- Built a scalable **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain** and **FAISS/Milvus**, accelerating document retrieval by **35%**.
- Deployed on **Azure Kubernetes Service** with health checks, auto-scaling, and secure role-based access ensuring continuous uptime.

AI Voice Agent for Policy-Bound Customer Support (VAPI + n8n) July 2025 – Aug 2025

- Built an **AI-driven voice support agent** using **VAPI** integrated with **n8n** workflows to handle inbound customer calls.
- Designed **policy-restricted response logic** ensuring the agent answers only within approved business rules and redirects unsupported queries.
- Implemented **call flow orchestration**, fallback handling, and escalation paths to human agents for out-of-scope requests.
- Logged call metadata and agent responses for **monitoring, audits, and continuous policy tuning**.

Real-Time Fraud Detection System Aug 2025 – Sep 2025

- Deployed using **AWS SageMaker** and **scikit-learn**, achieving sub-500ms latency for real-time **risk detection**.
- Containerized with **Docker** and automated CI/CD via **GitHub Actions**, ensuring rapid deployment in production.
- Monitored drift and anomalies with **CloudWatch**, enabling proactive fraud alerts for financial compliance.

Malaria Blood Cell Classification Using ResNet-50 Sep 2025 – Oct 2025

- Built a **ResNet-50** image classifier (TensorFlow/PyTorch) to detect malaria-infected blood cells with high accuracy.
- Performed **EDA, augmentation, imbalance handling**, and optimized training using learning-rate scheduling and regularization.
- Evaluated using **precision, recall, F1, ROC-AUC**, and applied **Grad-CAM** for explainability, aligning with healthcare requirements.

Tech Stack

- Cloud & MLOps:** AWS SageMaker, GCP Vertex AI, Azure AKS, Kubeflow, GitOps, Terraform, Docker, CI/CD, CloudWatch, Prometheus, Grafana
- AI/ML Frameworks:** Scikit-Learn, PyTorch, TensorFlow, LangChain, Hugging Face, LoRA/PEFT, RLHF, SHAP, LIME, OpenAI API
- Programming & Databases:** Python, SQL, C/C++, JavaScript (React, Node.js), FAISS, Milvus
- Big Data:** PySpark, Hadoop, Kafka, Data Warehousing, ELT/ETL Workflows
- Visualization & Tools:** Pandas, NumPy, Tableau, Grafana, Streamlit
- AI Engineering:** RAG, Multi-Agent Systems, Prompt Engineering, LLM Fine-Tuning
- RPA & Automation:** n8n, VAPI, workflow orchestration, rule-based process automation, API-triggered actions

Certifications

- Oracle** AI Vector Search Certified Professional
- Oracle Cloud Infrastructure** 2025 Certified Data Science Professional
- Oracle Cloud Infrastructure** 2025 Certified Generative AI Professional

Achievements

- Winner – HSBC Hackathon 2024 (solo; selected from 4,500+ applicants).
- Best Presentation – Spring Startup Sprint 2024.
- Best Research Paper – GITAM Ideathon 2024.
- 3rd Place – UI/UX Hackathon, IIIT Hyderabad 2025.